Safwan Hasan

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Education

University of Washington, Seattle — Seattle, WA
B.S. Computer Science; GPA: 4.0/4.0 — *Expected Jun. 2028*

- **Relevant Coursework:** Computer Programming III (CSE 123), Multivariable Calculus, Linear Algebra, Statistics

Experience

Rove (YC W24) — Remote
Software Engineering Intern — *Sep. 2024 – Dec. 2024*

- **Cut search latency 35%** by profiling slow queries and rewriting the hot path around targeted indexes, query batching, and a multi-layer cache
- Built and ran the Python/PostgreSQL search backend for a credit-card points platform handling **1,000+ flight queries per day across 10k+ records**; removed redundant upstream API calls from the data pipeline
- Shipped 15+ features to production with a 5-engineer team, planning scope with 3+ stakeholders and reviewing teammates' pull requests

Mobile Intelligence Lab, UW Allen School — Seattle, WA
AI Research Intern — *Jul. 2025 – Sep. 2025*

- Co-annotated the human pause ground truth for a Hibiki-based speech-to-speech translation model serving users with Cerebral Palsy, ALS, and Parkinson's, labeling **233 samples** from the **86.4-hour Speech Accessibility Project corpus**, then scored **596 Silero VAD detections** against it to trace pause failures to a 1.2s silence floor tuned on typical speech
- Retuning the floor to 400ms raised **time-level overlap from 42.2% to 84.6%** and segment overlap from 93.5% to 97.2%; validated the correction against Parkinson's and ALS speech to confirm it generalized beyond the condition it was tuned on

Stanford AIMI Center — Stanford, CA
Summer Research Intern — *Jun. 2024 – Jul. 2024*

- Extracted training labels from **12,716 radiologist reports** using RadGraph and spaCy, building the dataset that trained a chest X-ray classifier to 0.383 macro F1 across 4 disease classes under 30:1 class imbalance
- Built the NLP labeling layer of a 7-person end-to-end pipeline; system won 1st place among intern teams at final presentations

Projects

Roots | TypeScript, Next.js, AWS Lambda, DynamoDB, SQS, Claude API, Vitest — 2026

- Launched an internship platform that aggregates **1,000+ live listings** from community GitHub repos and tailors a LaTeX resume to each posting through a **6-stage Claude pipeline**, running serverless on AWS Lambda, SQS, EventBridge, and DynamoDB
- Verified every AI rewrite with LaTeX syntax checks, per-bullet diffs, and tokenizer-based keyword scoring instead of trusting model output, backed by **420 Vitest tests** gated on every push in GitHub Actions

PotLaunch | TypeScript, Next.js, PostgreSQL, Stripe Connect, PL/pgSQL — Spring 2026

- Engineered a crowdfunding platform where backers earn a share of a startup's revenue instead of rewards, using a Stripe Connect revenue oracle that distributes profit-sharing payouts automatically
- Enforced an immutable investment ledger with an append-only PostgreSQL table, mutation-blocking triggers, and **48 row-level security policies**, settling each payout in one PL/pgSQL transaction; took 2nd place at Niyyah Hacks

Setly | TypeScript, Next.js, PostgreSQL, Gemini API, SerpApi, Stripe Issuing — DubHacks 2026

- Built a group-checkout platform that generates multi-store product plans from natural-language prompts, validating Gemini output against Zod schemas, grounding prices in SerpApi data, and splitting costs in integer cents via Stripe
- Issued each funded group a Stripe Issuing virtual card restricted to the plan's stores, approving every charge through a real-time authorization webhook within a 2s budget

STEMLINK | Next.js, TypeScript, Firebase, Tailwind CSS, OpenAI API — 2025

- Developed a student job portal with an OpenAI-powered application assistant and role-based dashboards for students, employers, and admins, built on Next.js and Firebase; qualified for FBLA Nationals

Technical Skills

Languages: Java, Python, C++, C#/.NET, TypeScript, JavaScript, SQL, HTML/CSS
Databases & Cloud: PostgreSQL, DynamoDB, Firestore, PL/pgSQL, pgvector, TimescaleDB, Drizzle ORM, AWS (Lambda, SQS, EventBridge, S3, CloudWatch), Node.js, Express, Docker, Stripe API
ML / Scientific Computing: PyTorch, spaCy, RadGraph, scikit-learn, NumPy, pandas
Web & Tooling: React, Next.js, Tailwind CSS, Vitest, GitHub Actions CI/CD, Git